

Oxygen storage in muscle cells is mediated by the protein myoglobin, which consists of multiple α -helices held in a specific conformation. Each myoglobin protein binds to one heme cofactor, which in turn can bind one oxygen molecule. The bound oxygen can be released when a muscle cell's oxygen needs are not met by the blood. The oxygen binding curve for myoglobin is shown in Figure 1, with the p_{50} (the oxygen partial pressure at which 50% of myoglobin molecules are bound to oxygen) value shown.

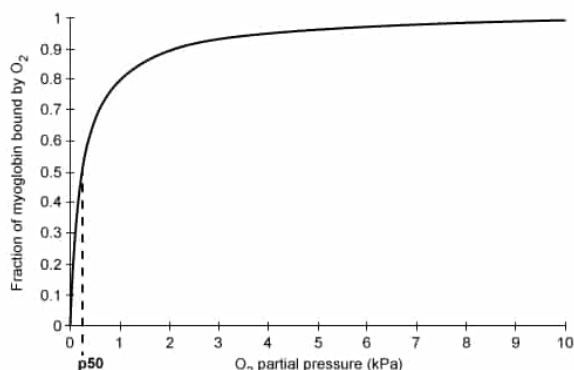


Figure 1 Oxygen binding curve of myoglobin

Myoglobin samples were exposed to different concentrations of the detergent *N,N*-dimethyl-*N*-dodecylglycine betaine (EBB).

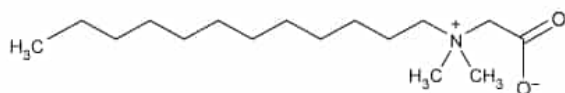


Figure 2 Structure of *N,N*-dimethyl-*N*-dodecylglycine betaine (EBB).

Oxygenated heme absorbs light at 409 nm when bound to myoglobin. This absorbance shifts slightly when heme is not bound to myoglobin. Figure 3 shows that as the concentration of EBB increases under atmospheric oxygen levels, the absorbance of the solution at 409 nm decreases, indicating that EBB causes partial myoglobin unfolding and heme release.

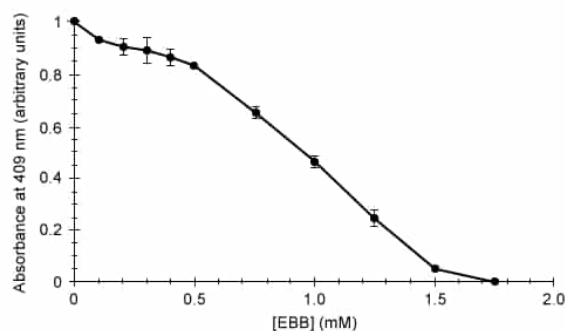


Figure 3 Absorbance of a myoglobin solution at 409 nm as a function of EBB concentration

Circular dichroism revealed that the α -helices within myoglobin remain largely intact at all EBB concentrations. Thermodynamic studies revealed that the standard Gibbs free energy of heme dissociation from myoglobin is approximately 8,000 kJ/mol.

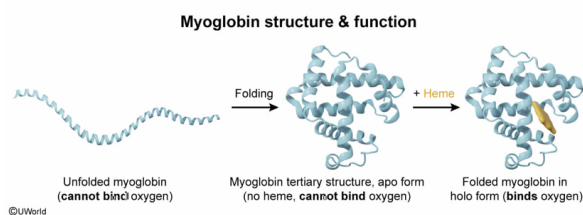
For myoglobin to bind oxygen, which of the following must be true?

- I. The partial pressure of oxygen must be above 0.25 kPa.
 - II. Myoglobin must have an intact tertiary structure.
 - III. Myoglobin must exist as a holoprotein.
- ☐ A. I and II only [14%]
☐ B. I and III only [9%]
☒ C. II and III only [61%]
☐ D. I, II, and III [13%]

Correct

61% answered correctly

Explanation:



For a protein to perform its biological function, it must be correctly folded into its three-dimensional form. The correct, folded form of a protein is known as its **tertiary structure**. **Unfolding or improper folding** (ie, disruption of tertiary structure) **prevents proper function**, sometimes through loss of necessary **cofactors** (even properly folded proteins that **lack their cofactors** are **unable to function**). A **correctly folded protein** that **contains all cofactors** is referred to as the **holoprotein** whereas those that **lack cofactors** are called **apoproteins**.

The biological function of myoglobin is to bind oxygen; therefore, binding oxygen requires myoglobin to have an **intact tertiary structure (Number II)**. However, oxygen does not bind to the amino acids of myoglobin themselves. Instead, it binds to the heme cofactor that is present in the holoform of myoglobin. Therefore, myoglobin **must exist as a holoprotein** (as opposed to an apoprotein) **to bind oxygen (Number III)**.

(Number I) Figure 1 shows that the p_{50} of oxygen for myoglobin binding is approximately 0.25 kPa. At this oxygen partial pressure, approximately 50% of the myoglobin molecules in a solution are bound to oxygen. At nonzero partial pressures

below this level, less than 50% but more than 0% of myoglobin molecules are bound.

Therefore, an oxygen partial pressure greater than 0.25 kPa is *not* required for at least some myoglobin molecules in a system to bind oxygen.

Educational objective:

The biologically active, three-dimensional, folded form of a protein is known as its tertiary structure. Disrupting tertiary structure interferes with a protein's ability to function. In addition, many proteins require cofactors to function properly, and a protein bound to its cofactor is known as a holoprotein.

Time spent:0 sec

Subject:
Biochemistry

QID:403474

System:
Amino Acids and Proteins

Oxygen storage in muscle cells is mediated by the protein myoglobin, which consists of multiple α -helices held in a specific conformation. Each myoglobin protein binds to one heme cofactor, which in turn can bind one oxygen molecule. The bound oxygen can be released when a muscle cell's oxygen needs are not met by the blood. The oxygen binding curve for myoglobin is shown in Figure 1, with the p_{50} (the oxygen partial pressure at which 50% of myoglobin molecules are bound to oxygen) value shown.

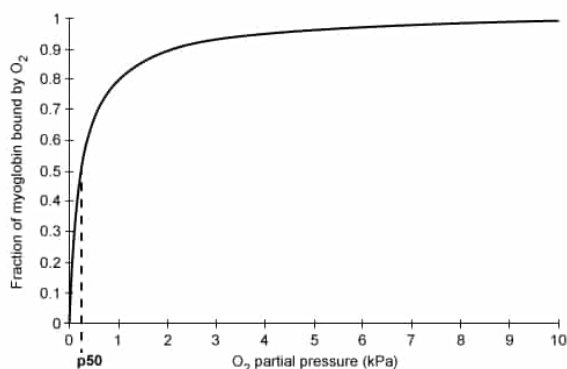


Figure 1 Oxygen binding curve of myoglobin

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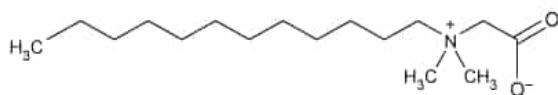


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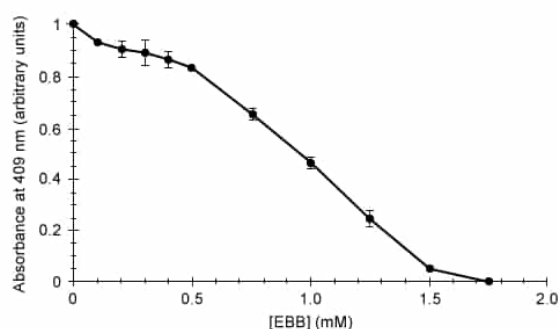
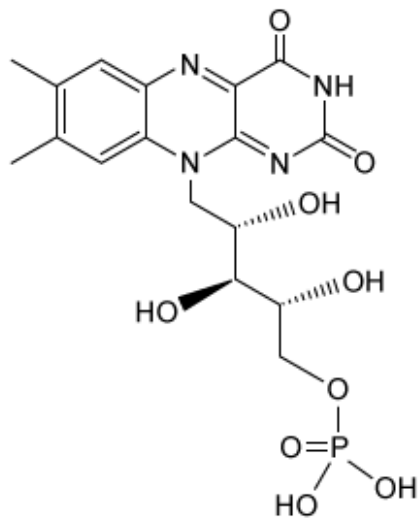


Figure 3 Absorbance of a myoglobin solution at 409 nm as a function of EBB concentration

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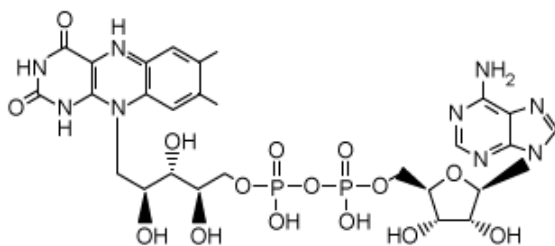
Which of the following best depicts the structure of the molecule released by myoglobin when EBB is added?

☐ A.



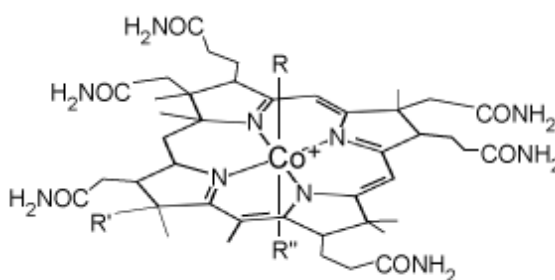
[8%]

☐ B.



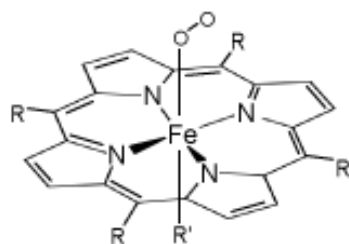
[5%]

☐ C.



[6%]

☒ D.

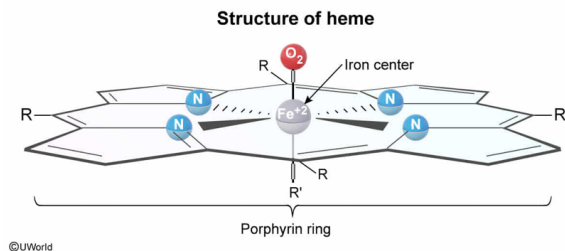


[78%]

Correct

78% answered correctly

Explanation:



Heme is a common **cofactor** found in many proteins, and it is required for oxygen binding in hemoglobin and myoglobin. The characteristic structure of heme is a **porphyrin ring** with a **central iron (Fe) atom**.

The passage states that myoglobin contains heme and that when EBB is added, myoglobin releases its heme cofactor. Of the given options, **only Choice D shows a porphyrin ring with an iron center**. Therefore, this diagram must depict the structure of the cofactor released when EBB is added to myoglobin.

(Choice A) This is the structure of flavin mononucleotide (FMN), which functions as a cofactor in various enzymes that catalyze electron transfer reactions. Myoglobin does not contain FMN.

(Choice B) This is the structure of flavin adenine dinucleotide (FADH₂). It is an important cofactor in the **electron transport chain** and other metabolic pathways that transfer electrons, but it is not found in myoglobin.

(Choice C) This is the structure of vitamin B₁₂. Although it is similar in structure to heme, it contains several differences. Most notably, vitamin B₁₂ contains a cobalt (Co) center rather than an iron (Fe) center.

Educational objective:

Heme is a cofactor derived from porphyrin rings. It is found in many proteins and commonly helps bind oxygen gas, which interacts with the iron center of the heme molecule.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403475

System:
Amino Acids and Proteins

Oxygen storage in muscle cells is mediated by the protein myoglobin, which consists of multiple α -helices held in a specific conformation. Each myoglobin protein binds to one heme cofactor, which in turn can bind one oxygen molecule. The bound oxygen can be released when a muscle cell's oxygen needs are not met by the blood. The oxygen binding curve for myoglobin is shown in Figure 1, with the p_{50} (the oxygen partial pressure at which 50% of myoglobin molecules are bound to oxygen) value shown.

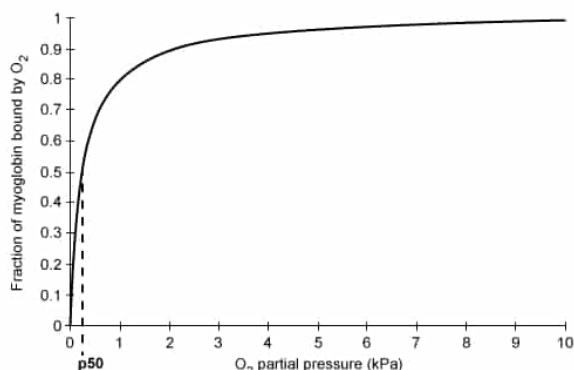


Figure 1 Oxygen binding curve of myoglobin

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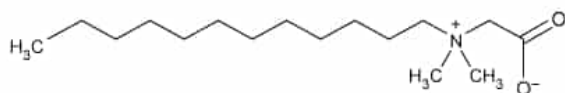


Figure 2 Structure of *N,N*-dimethyl-*N*-dodecylglycine betaine (EBB).

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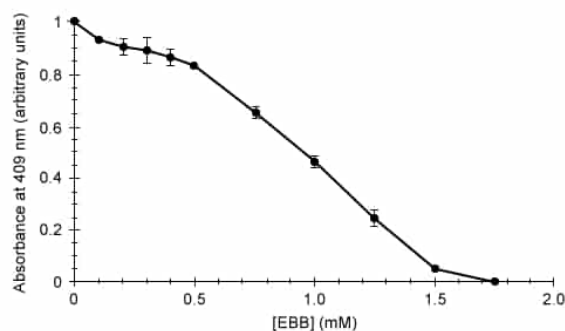


Figure 3 Absorbance of a myoglobin solution at 409 nm as a function of EBB concentration

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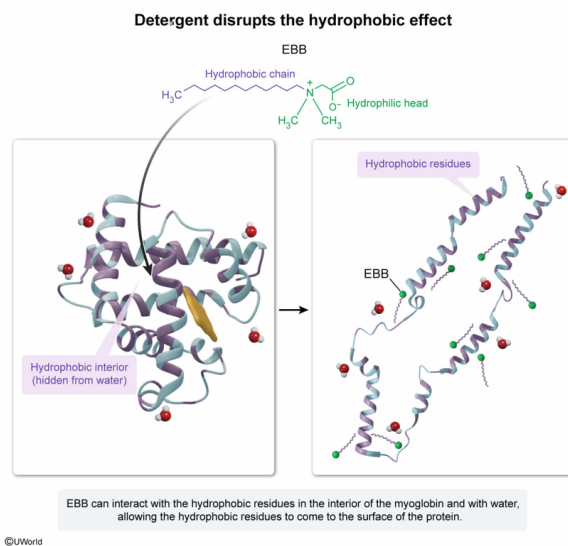
EBB most likely causes protein unfolding by:

- ☐ A. disrupting hydrogen bonds that stabilize the α -helices of myoglobin. [25%]
- ✓ ☒ B. disrupting hydrophobic interactions in the interior of myoglobin. [36%]
- ☐ C. disrupting ionic interactions in the interior of myoglobin. [19%]
- ☐ D. disrupting hydrophobic interactions between side chains of surface α -helices. [18%]

Correct

37% answered correctly

Explanation:



The **tertiary structure** of a protein is **stabilized by multiple forces**, including ionic bonds and hydrogen bonds between hydrophilic surface side chains as well as hydrophobic interactions between side chains in the protein core. Of these forces, the hydrophobic interactions play the dominant role in driving protein folding. Likewise, protein **denaturation** (unfolding) is commonly driven by amphipathic molecules (ie, molecules containing both hydrophilic and hydrophobic regions) that **disrupt the energetically favorable hydrophobic interactions** between amino acids in the interior of the protein.

This is due to the **hydrophobic effect**: In the absence of denaturant molecules, the **"hiding" of hydrophobic residues in the interior** of the protein allows **water molecules surrounding the protein** to adopt a much **higher entropy state**, resulting in an overall large, negative Gibbs free energy of folding. In contrast, formation of hydrogen bonds between two polar side chains is at best only slightly more energetically favorable than those same side chains forming hydrogen bonds with water.

Consequently, although such hydrogen bonds may help *stabilize* folded protein structure, they are *not* the primary factor in protein folding itself.

The structure of EBB shown in Figure 2 depicts a long hydrocarbon chain that could easily interact with hydrophobic residues in proteins, disrupting the interactions between these residues in the protein core. In addition, the hydrophilic end of EBB can interact well with water, allowing the hydrophobic residues inside the protein to be brought to the surface. Therefore, **EBB most likely causes protein unfolding by disrupting hydrophobic interactions in the interior of myoglobin.**

(Choices A and D) The passage states that **α -helices** remain largely intact in the presence of EBB. Therefore, EBB most likely does not disrupt many interactions within α -helices. In addition, *hydrophobic* side chains are unlikely to be found in α -helices at the surface of the folded protein, where they would interact poorly with water.

(Choice C) The interior of a protein is unlikely to contain many ionic interactions because the residues in the inside of a protein are predominantly hydrophobic.

Educational objective:

Chemical denaturation of proteins is commonly caused by the disruption of hydrophobic interactions in the interior of the protein.

Time spent:0 sec

Subject:
Biochemistry

QID:403476

System:
Amino Acids and Proteins

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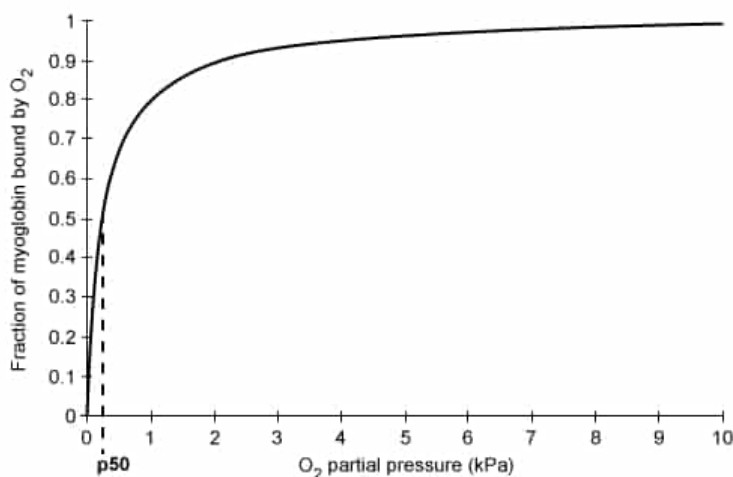


Figure 1 Oxygen binding curve of myoglobin

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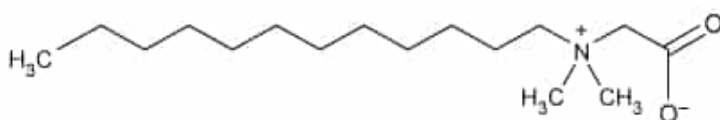


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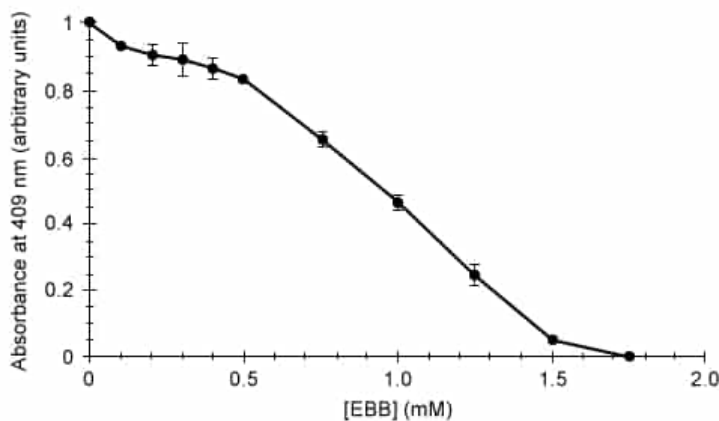


Figure 3 Absorbance of a myoglobin solution at 409 nm as a function of EBB concentration

Circular dichroism revealed that the α -helices within myoglobin remain largely intact at all EBB concentrations. Thermodynamic studies revealed that the standard Gibbs free energy of heme dissociation from myoglobin is approximately 8,000 kJ/mol.

Based on the free energy of dissociation of the heme-myoglobin complex, within what range does the K_d of the heme-myoglobin binding interaction fall?

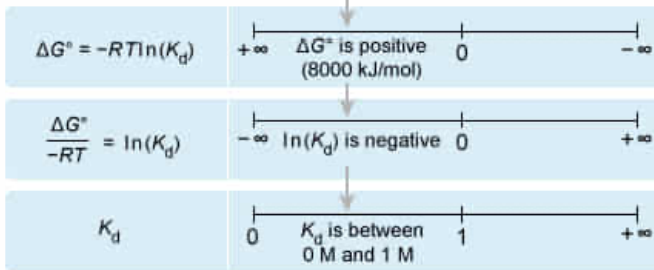
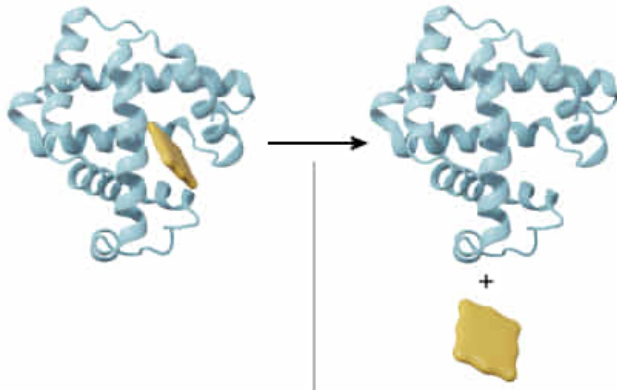
- ✓ ☒ A. Between 0 M and 1 M [50%]
- ☐ B. Equal to 1 M [5%]
- ☐ C. Greater than 1 M [22%]
- ☐ D. The range of possible K_d values cannot be determined without more information. [21%]

Correct

51% answered correctly

Explanation:

Free energy of dissociation and K_d



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The **dissociation constant** K_d of a binding interaction is the **equilibrium constant** for the dissociation of the two binding partners (commonly a protein and its ligand). The standard units for K_d are molar (M), although for convenience K_d values are sometimes converted to mM or μM . A low K_d indicates a strong binding interaction and a high K_d indicates a weak interaction.

The equilibrium constant for any process is related to the **standard Gibbs free energy** change (ΔG°) of that process by the equation:

$$\Delta G^\circ = -RT \ln(K)$$

where R is the universal gas constant, T is the **Kelvin temperature** of the system, and K is the equilibrium constant. This equation can be rearranged as:

$$\frac{\Delta G^\circ}{-RT} = \ln(K)$$

Based on this equation, when ΔG° is a positive number $\ln(K)$ is a negative number, because $\ln(K)$ is isolated by dividing ΔG° by a negative number. In this scenario, K must be between 0 and 1

because the logarithm of any number between 0 and 1 is negative whereas the logarithm of any number greater than 1 is positive. Because the reported ΔG° of myoglobin dissociation from heme is a positive number (8,000 kJ/mol), the related equilibrium constant (K_d) must be between 0 M and 1 M.

(Choice B) A K_d value of 1 M would only occur if ΔG° of dissociation were exactly 0 kJ/mol (ie, $\ln(1) = 0$).

(Choice C) A K_d value greater than 1 M would occur if ΔG° of dissociation were *negative*.

(Choice D) Based on the mathematical relation between ΔG° and the equilibrium constant of any process, knowing the ΔG° of dissociation provides enough information to determine whether the dissociation constant K_d is greater than, equal to, or less than 1. (It is physically impossible to have an equilibrium constant less than 0.)

Educational objective:

The standard Gibbs free energy ΔG° of a process is related to its equilibrium constant K by the equation

$$\Delta G^\circ = -RT\ln(K)$$

where R is the universal gas constant and T is the Kelvin temperature. A negative ΔG° always indicates an equilibrium constant greater than 1, and a positive ΔG° always indicates an equilibrium constant between 0 and 1.

Time spent:0 sec
Subject:
Biochemistry

QID:403477
System:
Amino Acids and Proteins